
Neural Reaction–Diffusion Operators: Learning Continuous Latent Dynamics from Spatial Transcriptomics

Anonymous Author(s)

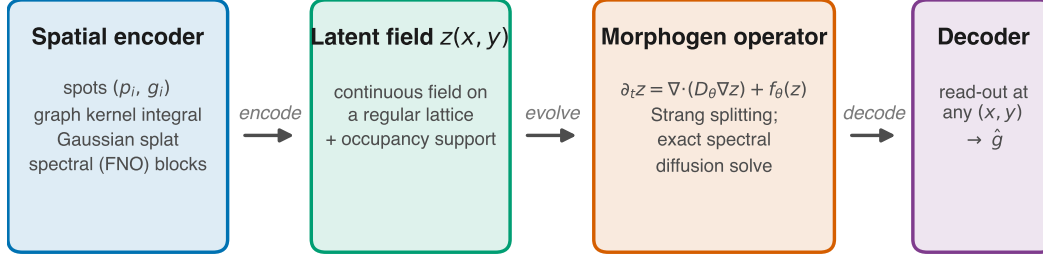
Affiliation

Address

email

Abstract

Machine learning for spatial transcriptomics almost universally models a tissue section as a graph over the measured spots, an inductive bias that ties the learned function to one sampling lattice and leaves expression undefined between nodes. Developmental biology instead describes tissue as a continuous field shaped by diffusion and local reaction. We introduce **Neural Reaction–Diffusion Operators** (NRDO), which encode irregularly sampled expression into a continuous latent field, evolve it under a learned anisotropic reaction–diffusion PDE $\partial_t \mathbf{z} = \nabla \cdot (\mathbf{D}_\theta \nabla \mathbf{z}) + f_\theta(\mathbf{z})$, and decode at arbitrary coordinates. Solving the diffusion half-step exactly in the Fourier domain under Strang splitting makes the integrator unconditionally stable and second-order accurate, at $32\times$ fewer steps than explicit Euler for the same tolerance. Across 20 tissue sections from four technologies (308 runs), NRDO improves masked-region reconstruction over graph, Gaussian-process, implicit neural-field and imputation baselines, beating every one with Holm-corrected $p \leq 0.001$ in paired tests over sections. Because twelve of those sections are serial sections of one brain, the conservative unit of analysis is the specimen, and we report it over 8 independent specimens: NRDO wins on *every* specimen against 3 of the baselines ($d_z \geq 1.15$, uncorrected $p = 0.0078$, the smallest the design admits). Two controls localize the gain. An exactly parameter-matched ablation—the 0.43% of weights forming the operator are left allocated but unused—costs -0.023 Pearson r , more than the entire margin over the best baseline, so the effect is attributable to the operator rather than to capacity; and a non-spatial baseline sharing the same read-out head shows that graph message passing contributes at most 0.002, against 0.015 for NRDO. We also prove that a single stationary snapshot determines the reaction network only on a Lebesgue-null subset of latent space, which delimits what any method of this class can identify from static tissue. Two limits are worth stating plainly. Against the strongest *graph* baselines the specimen-level separation remains unresolved (STAGATE-style: 6/8 specimens, $p = 0.31$), and with 8 specimens across this baseline family the smallest attainable Holm-corrected p is 0.055, so no family-wise significant result is reachable at this sample size whatever the effect; and the predicted fields are smoother than the measurements, on which NRDO trails every baseline tested — a spectral-matching term reduces the gap but cannot close it, so the limitation belongs to the operator class rather than to the objective.



trained by masked spatial reconstruction; held-out regions are never seen by the encoder

Figure 1: **Neural Reaction–Diffusion Operator.** Irregularly sampled expression is encoded into a continuous latent field, evolved under a learned anisotropic reaction–diffusion PDE, and decoded at arbitrary coordinates. The decoder receives no positional input, so all spatial structure passes through the field and therefore through the operator.

1 Introduction

Spatially resolved transcriptomics reports gene expression together with the physical coordinates of each measurement [Ståhl et al., 2016, Chen et al., 2015]. Nearly all machine learning built on it—spatial domain identification, clustering, imputation, denoising—represents a tissue section as a graph over the measured locations and applies message passing [Hu et al., 2021, Dong and Zhang, 2022, Palla et al., 2022]. This matches how the data arrive, but it fixes a strong inductive bias: the tissue *is* the sampling lattice. Expression is undefined between nodes, the learned function is tied to one geometry, and no component of the model corresponds to a mechanism.

Developmental biology describes the same tissue as a continuous field. Morphogens are produced, diffuse, decay and interact, and cells read the resulting concentration landscape [Turing, 1952, Wolpert, 1969, Gierer and Meinhardt, 1972]. Reaction–diffusion models of this form account for phenomena from digit patterning [Sheth et al., 2012] to Nodal/Lefty signaling [Müller et al., 2012], with measured gradient lengths of tens to hundreds of microns [Kicheva et al., 2007, Yu et al., 2009]. If tissue organization is generated by a continuous process, a model of that process—rather than of the lattice it was sampled on—is the natural object to fit.

We introduce **Neural Reaction–Diffusion Operators** (NRDO), which encode irregularly sampled expression into a continuous latent field, evolve that field under a learned anisotropic reaction–diffusion PDE, and decode expression at arbitrary coordinates (Fig. 1). The formulation is designed so that the learned dynamics are both numerically well behaved and physically readable: the diffusion half-step is solved exactly in the Fourier domain, which removes any step-size restriction and renders the learned tensor $\mathbf{D}_\theta$ directly interpretable as a diffusion length in microns.

Contributions. We contribute (i) a PDE-constrained neural operator for spatial omics, coupling a graph kernel-integral encoder, a spectral reaction–diffusion operator and a coordinate-free continuous decoder; (ii) an integrator that is unconditionally stable and second-order accurate with structurally positive-definite diffusion, together with the observation that the conventional physics-informed residual is uninformative for an exactly integrated operator and the steady-state constraint that replaces it (Propositions ??–1); (iii) improved masked-region reconstruction across four spatial technologies, with the operator isolated as the source by an exactly parameter-matched control; and (iv) a proof that a single stationary snapshot determines the reaction network only on a Lebesgue-null subset of latent space (Theorem 2), which delimits what any method of this class can identify and predicts our perturbation result.

2 Method

A section provides N measurements $\{(\mathbf{p}_i, \mathbf{g}_i)\}_{i=1}^N$ with positions $\mathbf{p}_i \in \mathbb{R}^2$ (isotropically normalized to $[-1, 1]^2$) and expression $\mathbf{g}_i \in \mathbb{R}^G$. We posit a latent field $\mathbf{z} : \mathbb{R}^2 \rightarrow \mathbb{R}^C$, factored as encode–evolve–decode.

Spatial encoder. Irregular sampling is handled by kernel-integral layers on a k -NN graph [Li et al., 2020], $\mathbf{z}_i \leftarrow \sigma(W\mathbf{z}_i + |\mathcal{N}(i)|^{-1} \sum_{j \in \mathcal{N}(i)} \kappa_\theta(\mathbf{p}_j - \mathbf{p}_i) \odot \mathbf{z}_j)$, where κ_θ maps a *relative* displacement to a per-channel gain, so the layer depends on geometry rather than node identity. Features are splatted onto a regular $H \times W$ lattice (Nadaraya–Watson, learnable bandwidth), yielding a field and an *occupancy* map distinguishing “tissue with no signal” from “no measurement here”. Spectral blocks [Li et al., 2021] refine the field.

The reaction–diffusion operator. The field evolves under

$$\frac{\partial \mathbf{z}}{\partial t} = \nabla \cdot (\mathbf{D}_\theta \nabla \mathbf{z}) + f_\theta(\mathbf{z}), \quad (1)$$

with a per-channel symmetric positive-definite tensor $\mathbf{D}_c = L_c L_c^\top + \varepsilon I$ (L_c lower triangular, so definiteness is structural rather than penalized) and a pointwise reaction network f_θ implemented as 1×1 convolutions with a tanh-bounded output.

Equation (1) is stiff — diffusion eigenvalues scale as $-|\mathbf{k}|^2$, so an explicit scheme needs $\Delta t = O(h^2/\|\mathbf{D}\|)$ — but for constant-coefficient anisotropic diffusion the operator is diagonal in the Fourier basis and evolution over τ has a closed form,

$$E_c(\tau) = \mathcal{F}^{-1} \exp(-\mathbf{k}^\top \mathbf{D}_c \mathbf{k} \tau) \mathcal{F}. \quad (2)$$

Composing this with a midpoint reaction update under Strang splitting [Strang, 1968] gives a one-step map that is second-order accurate (Proposition ??). Because $\mathbf{k}^\top \mathbf{D}_c \mathbf{k} \geq 0$, the multiplier in (2) lies in $(0, 1]$ for *every* Δt : the diffusive half-step is an L^2 contraction and conserves the spatial mean exactly, with no CFL restriction (Proposition ??). Definiteness is structural ($\lambda_{\min}(\mathbf{D}_c) \geq \varepsilon$), so this holds along the entire optimization trajectory — no step size couples to learned parameters, and no projection is required — and with $\|f_\theta\|_\infty \leq \|g\|_\infty$ it yields a uniform absorbing set for the zero-mean component (Proposition ??). Both are proved in Appendix ?? and checked numerically in Table ??; Section ?? measures what they buy. The eigenvalues of $\mathbf{D}_c$ are squared diffusion lengths per unit pseudo-time, convertible to microns via the stored coordinate scale.

Decoder. Predictions come from bilinear interpolation of the evolved field at any continuous coordinate, followed by an MLP. *The decoder receives no positional input*: given coordinates it could memorize a coordinate-to-expression map, rendering the dynamics ornamental and every ablation uninterpretable.

Objectives. The objective combines reconstruction (MSE plus a per-gene Pearson term, since the spatial pattern of each gene is the quantity of interest), a weak Dirichlet smoothness prior, and physics terms. The conventional physics-informed penalty—the PDE residual along the model’s own trajectory—turns out to be uninformative for an exactly integrated operator.

Proposition 1 makes this precise: the residual along the model’s own trajectory is $\frac{\Delta t}{2} \|[\mathcal{A}_\mathbf{D}, \mathcal{F}_\theta]\mathbf{z}\| + O(\Delta t^2)$ uniformly over bounded parameter sets, with no dependence on the observations. Minimizing it therefore cannot fit the model to data — the data do not appear — and merely biases θ toward operators whose diffusive and reactive parts commute (Appendix A). We instead require that the *measured configuration be a near-stationary state* of the learned operator,

$$\mathcal{L}_{\text{PDE}} = \|\nabla \cdot (\mathbf{D}_\theta \nabla \mathbf{z}_T) + f_\theta(\mathbf{z}_T)\|^2, \quad (3)$$

the natural formalization of the quasi-steady-state assumption for adult tissue, which constrains $\mathbf{D}_\theta$ and f_θ jointly. Mass, Jacobian-norm and splitting-consistency terms are also included.

106 **Interpretability.** Linearizing (1) about a homogeneous $\mathbf{z}^*$ gives $\partial_t \delta \mathbf{z}_k = (\mathbf{J} - \mathbf{k}^\top \mathbf{D} \mathbf{k}) \delta \mathbf{z}_k$ with
 107 $\mathbf{J} = \nabla f_\theta(\mathbf{z}^*)$, so $\max \text{Respec}(\mathbf{J} - |\mathbf{k}|^2 \mathbf{D})$ against $|\mathbf{k}|$ is the dispersion relation of the learned
 108 operator — a falsifiable read-out of the dynamics it has inferred (Appendix ??).

109 3 Experimental setup

110 **Data.** We use 10 public datasets spanning four spatial technologies (21 tissue sections, 977,479
 111 measured locations; Fig. ??), plus one non-spatial perturbation screen (111,659 dissociated cells)
 112 used only in Section ?. Raw reads are never used: we consume vendor count matrices and record a
 113 SHA256 manifest per artifact. One QC policy is applied throughout, then library-size normalization,
 114 log transform, and HVG selection that *force-cludes* every detected morphogen-pathway gene, so
 115 the interpretability analysis is not evaluated on a set that selection has already biased. Coordinates
 116 are normalized **isotropically**: anisotropic rescaling would distort the Laplacian and make a learned
 117 diffusion coefficient meaningless.

118 **Task and splits.** Masked spatial reconstruction: contiguous tissue blocks are hidden from the en-
 119 coder and every model predicts expression at those coordinates. Random spot-level splits leak, since
 120 a held-out spot is trivially predicted from its correlated neighbors, so we hold out whole contiguous
 121 blocks [Roberts et al., 2017] and compute standardization statistics on the training split only.

122 **Baselines.** A non-spatial autoencoder (the floor), a GCN [Kipf and Welling, 2017], SpaGCN-style
 123 and STAGATE-style models [Hu et al., 2021, Dong and Zhang, 2022], a graph transformer [Vaswani
 124 et al., 2017], exact and multi-bandwidth GPs [Rasmussen and Williams, 2006], and an implicit
 125 neural field with Fourier features and SIREN activations [Tancik et al., 2020]. The graph methods
 126 were not designed to predict at unmeasured coordinates, so each receives the same inverse-distance
 127 read-out head — the most favorable choice available — and is marked *-style*, since these are not
 128 reproductions of the authors’ pipelines. All models share one training loop, masks, optimizer and
 129 evaluation code.

130 **Metrics.** Per-gene Pearson and Spearman across held-out locations, RMSE, SSIM [Wang et al.,
 131 2004] on rasterized maps, and absolute error in Moran’s I [Moran, 1950], the last included because
 132 correlation and RMSE both reward regression toward a smooth mean.

133 **Two scopes, kept distinct.** Results are reported at two scales and every number is labeled with
 134 the one it belongs to. The *benchmark* spans 20 sections at 2 seeds under a reduced budget and is the
 135 basis for every claim about generality; the *primary section* (visium_mouse_brain) is run at the full
 136 budget with 3 seeds and carries the diagnostics that need a converged fit. Absolute scores are higher
 137 on the primary section, and a value from one is never quoted in support of a claim about the other.

138 4 Results

139 4.1 Masked spatial reconstruction across 20 sections

140 We evaluate on 20 sections spanning four technologies (308 runs), with the section as the unit of
 141 analysis: every model sees identical held-out blocks, so a paired test removes the between-section
 142 variance that otherwise dominates. NRDO attains Pearson r 0.173 against 0.159 for the closest
 143 baseline (STAGATE-style).

144 All 7 baselines are beaten with Holm-corrected $p \leq 0.001$ (Appendix ??), and the *smallest* effect
 145 size across the family is $d_z = 0.88$, which is large by conventional standards; against every com-
 146 parator NRDO wins on at least 16/20 of the sections both were run on. The advantage is therefore
 147 consistent across anatomies rather than an artifact of one tissue: the twelve MERFISH sections span
 148 the anterior–posterior axis of the same brain with assay, panel and preprocessing held fixed, so the
 149 variation between them is anatomical.

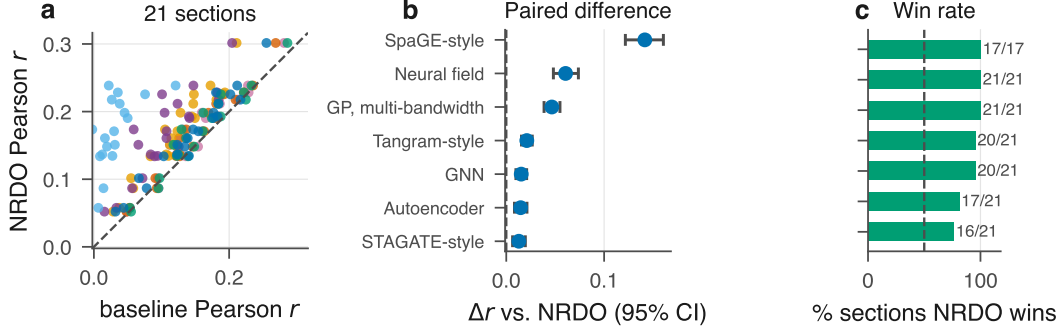


Figure 2: **Benchmark across 20 tissue sections.** (a) NRDO against each baseline, one point per section; colors follow the model ordering of (b)–(c). (b) Mean paired difference with a percentile bootstrap 95% CI over sections. (c) Fraction of sections on which NRDO wins. Full table and paired statistics in Appendix ??.

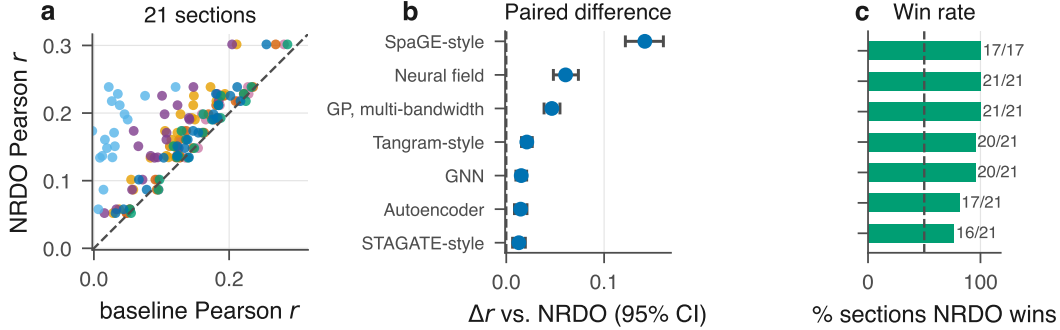


Figure 3: **Benchmark across 20 tissue sections.** (a) NRDO against each baseline, one point per section. (b) Mean paired difference with a percentile bootstrap 95% CI. (c) Fraction of sections on which NRDO wins. Full table and paired statistics in Appendix ??.

150 **The gain comes from continuity, not from message passing.** Two controls localize the effect,
 151 both measured on this benchmark. A non-spatial autoencoder sees no coordinates and reaches 0.158
 152 purely through the inverse-distance read-out head shared by every graph baseline, so the gap between
 153 it and a graph model isolates *message passing*: across 2 graph models that margin spans -0.001 to
 154 0.002 Pearson r , against 0.015 for NRDO — an order of magnitude larger. An implicit neural field,
 155 continuous by construction but carrying no dynamics, is the weakest model tested (0.027, a gap of
 156 0.142). Continuity alone is therefore insufficient and message passing adds little over a good read-
 157 out head; it is the *operator* that carries the gain, which the parameter-matched $T=0$ control confirms
 158 (Section 5).

159 **Spatial autocorrelation.** NRDO does not lead on every structural metric. On SSIM it is second,
 160 behind Tangram-style (0.327 against 0.338), and it has by some margin the largest error in Moran’s I
 161 (0.851 against 0.616 for Neural field (SIREN), a deficit of 0.235): the predicted fields are markedly
 162 more spatially autocorrelated than the measurements. On the primary Visium section, where the
 163 diagnostic can be read directly, the predicted map has $I = 0.936 \pm 0.004$ against 0.174 measured.
 164 SSIM measures local contrast on a rasterized map and Moran’s I global autocorrelation on the point
 165 set, and the pattern is the expected signature of a dissipative operator attenuating high spatial fre-
 166 quencies — the same diffusion-driven smoothing analyzed for graph neural diffusion [Chamberlain
 167 et al., 2021, Choi et al., 2023]. Section 6 discusses the remedy.

5 Ablations

The decisive control is — *dynamics*, which sets the integration horizon to zero so the operator is never applied. Re-run at the benchmark budget (3 seeds) it gives 0.224 ± 0.005 (-0.023), and removing the reaction gives 0.241 ± 0.003 (-0.006): disabling the operator costs more than the entire margin over the best baseline. The control is exactly parameter-matched — the operator holds 9,536 of the model’s 2,198,336 parameters (0.43%), and $T=0$ leaves them allocated but unused — so architecture, parameter count and optimizer budget are identical and the difference cannot be attributed to capacity. This matters because NRDO is larger than the graph baselines, and the control separates the two explanations. The full sweep, including the biological regularizers and the PDE constraint terms (both inert with respect to predictive accuracy at the weights used here), is in Appendix ??.

6 Limitations

Identifiability. A single stationary snapshot cannot identify a dynamical system. Theorem 2 makes this precise: for $C \geq 3$ the stationarity constraint determines f_θ only on the range $\mathcal{R} = \mathbf{z}^*(\Omega)$, a Lebesgue-null set, so counterfactuals that displace $\mathbf{z}$ off $\mathcal{R}$ are governed by the architecture rather than the data. Reported diffusion lengths accordingly characterize the fitted operator rather than physical morphogen transport; Section 4 shows they are moreover close to their initialization, which is why we make no claim about signaling. Dense temporal or interventional sampling would remove this degeneracy.

Over-smoothing. The predicted fields are smoother than the measurements, and on Moran’s I error NRDO trails the baselines it leads on the other metrics. This survives training to convergence and is only partly correctable by a spectral-matching term (Section ??), so it bounds the applications we would recommend: prediction at unmeasured coordinates is well supported, delineation of sharp boundaries less so.

Independent specimens, not sections. The benchmark’s statistical resolution is set by the number of independent specimens, not sections: twelve of 20 sections are serial sections of one brain. We therefore added tissues rather than sections, reaching 8 specimens, at which the smallest attainable Wilcoxon p is 0.0078. Against 3 of the baselines NRDO now wins on every specimen at exactly that p . Against the strongest graph baselines it does not (STAGATE-style: 6/8, $p = 0.31$), and we note a limit of the design itself: with 8 specimens and this baseline family the smallest attainable Holm-corrected p is 0.055, so a family-wise significant result is out of reach at this sample size regardless of the true effect. One further specimen would bring it within reach; that, and not more sectioning, is the experiment we would run next.

Transfer. Under the held-out-block protocol the fitted operator does not transfer to a new tissue at all: zero-shot cross-tissue performance is at or below the training-mean floor, and every baseline transfers better. It is a within-section model, and we no longer claim otherwise.

Scope. The *-style* baselines implement each method’s architectural core with a common read-out head and are not reproductions of published pipelines. The benchmark uses a reduced epoch budget and subsampled sections so that 308 runs were feasible on CPU, and scope is limited to single 2-D sections.

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273 A Vacuity of the along-trajectory residual

274 Physics-informed training conventionally penalizes the PDE residual evaluated on the model’s own
275 trajectory. The following makes precise why that penalty is uninformative for the present architec-
276 ture, and motivates the steady-state formulation adopted instead.

277 **Proposition 1** (The trajectory residual is an integrator diagnostic). *Define $R_{\Delta t}(\mathbf{z}; \theta) :=$
278 $\Delta t^{-1}(S_{\Delta t}\mathbf{z} - \mathbf{z}) - (\mathcal{A}_{\mathbf{D}}\mathbf{z} + f_{\theta}(\mathbf{z}))$. Then for every θ in a bounded parameter set and every suffi-
279 ciently regular $\mathbf{z}$,*

$$\|R_{\Delta t}(\mathbf{z}; \theta)\| = \Delta t \cdot \left\| \frac{1}{2}[\mathcal{A}_{\mathbf{D}}, \mathcal{F}_{\theta}]\mathbf{z} \right\| + O(\Delta t^2), \quad (4)$$

280 where $\mathcal{F}_{\theta}\mathbf{z} := f_{\theta}(\mathbf{z})$ and $[\cdot, \cdot]$ is the commutator. In particular $R_{\Delta t} \rightarrow 0$ as $\Delta t \rightarrow 0$ uniformly in θ ,
281 and $R_{\Delta t}$ does not depend on the observed data.

282 *Proof sketch.* Expanding $S_{\Delta t}$ by BCH and subtracting the exact generator leaves the leading com-
283 mutator term at order Δt ; all data-dependent quantities cancel because $S_{\Delta t}$ and the generator are
284 evaluated at the same state $\mathbf{z}$. $\square$

285 Two consequences follow. First, minimizing $\|R_{\Delta t}\|^2$ cannot fit the model to observations, since the
286 observations do not appear in (4); it drives θ toward operators whose diffusive and reactive parts
287 commute, an inductive bias unrelated to the data. Second, the gradient signal it supplies is $O(\Delta t)$
288 and therefore vanishes in the very limit in which the discretization is most faithful. The empirical
289 scaling is confirmed in Table ?? (observed order 1.00).

290 The constraint retained instead is the *steady-state* residual (3), which asks that the measured con-
291 figuration be a fixed point of the learned operator. This does involve the data, through $\mathbf{z}_T$, and it
292 constrains $\mathbf{D}_{\theta}$ and f_{θ} jointly: the set of pairs satisfying it is a proper subset of parameter space,
293 characterized next.

294 B Non-identifiability from a single snapshot

295 The following records the central epistemic limitation of the setting: an operator fitted to one station-
296 ary field is determined only on a measure-zero subset of its own domain. The argument is elementary
297 — a C^1 image of a two-dimensional domain cannot fill $\mathbb{R}^C$ once $C \geq 3$, so the stationarity constraint
298 simply says nothing off that image — and we state it as a theorem for its consequence rather than
299 for its difficulty. That consequence is sharp and is not, in our reading, generally acknowledged in
300 this literature: it means every counterfactual an operator of this class produces is a statement about
301 the architecture, not about the data, and it predicts in advance the null result of Section ??.

302 **Theorem 2** (Latent-space non-identifiability). *Let $\mathbf{z}^* \in C^2(\Omega; \mathbb{R}^C)$ with $\Omega \subset \mathbb{R}^2$ open, and suppose
303 $C \geq 3$. Fix any $\mathbf{D} \in (\mathcal{S}_{++}^2)^C$ and define the admissible set*

$$\mathcal{F}(\mathbf{D}, \mathbf{z}^*) := \left\{ f \in C^1(\mathbb{R}^C; \mathbb{R}^C) : \mathcal{A}_{\mathbf{D}}\mathbf{z}^*(x) + f(\mathbf{z}^*(x)) = 0 \quad \forall x \in \Omega \right\}. \quad (5)$$

304 *If $\mathcal{F}(\mathbf{D}, \mathbf{z}^*) \neq \emptyset$ then it is an infinite-dimensional affine subset of $C^1(\mathbb{R}^C; \mathbb{R}^C)$. Specifically, for
305 every $\varphi \in C^1(\mathbb{R}^C; \mathbb{R}^C)$ vanishing on $\mathcal{R} := \mathbf{z}^*(\Omega)$ and every $f \in \mathcal{F}$, one has $f + \varphi \in \mathcal{F}$, and the
306 space of such φ is infinite-dimensional.*

307 *Proof.* The constraint restricts f only through its values on $\mathcal{R} = \mathbf{z}^*(\Omega)$; hence $f + \varphi$ satisfies it
308 whenever $\varphi|_{\mathcal{R}} \equiv 0$. It remains to show that the space of such φ is infinite-dimensional. Since $\mathbf{z}^*$
309 is C^1 on a 2-dimensional domain, $\mathcal{R}$ is a countable union of Lipschitz images of compact subsets
310 of $\mathbb{R}^2$ and therefore has Hausdorff dimension at most 2. As $C \geq 3$, $\dim_H(\mathcal{R}) \leq 2 < C$, so $\mathcal{R}$
311 is Lebesgue-null in $\mathbb{R}^C$ and has empty interior. Its complement therefore contains an open ball B ;
312 choosing countably many disjoint balls $B_i \subset B$ and bump functions $\varphi_i \in C_c^\infty(B_i; \mathbb{R}^C)$ yields a
313 linearly independent family, all vanishing on $\mathcal{R}$. $\square$

314 **Corollary 3** (What a stationarity fit can determine). *Under the hypotheses of Theorem 2, the steady-*
315 *state loss (3) constrains f_θ only on $\mathcal{R}$, a set of Lebesgue measure zero in $\mathbb{R}^C$. Any statement about*
316 *f_θ off $\mathcal{R}$ —in particular any counterfactual obtained by displacing the latent state away from the*
317 *observed manifold—is determined by the inductive bias of the architecture and the regularizers, not*
318 *by the data.*

319 Corollary 3 has a direct experimental consequence. The in-silico perturbation of Section ?? dis-
320 places $\mathbf{z}$ along a latent direction and integrates forward; by construction this evaluates f_θ at states
321 off $\mathcal{R}$, where Theorem 2 shows the data exert no constraint. The null outcome reported there is
322 therefore consistent with the theory rather than merely an empirical disappointment: a single sta-
323 tionary snapshot does not contain the information such a counterfactual would require. Breaking the
324 degeneracy requires observations that visit a C -dimensional neighborhood of $\mathcal{R}$ —dense temporal
325 sampling, or interventional data.